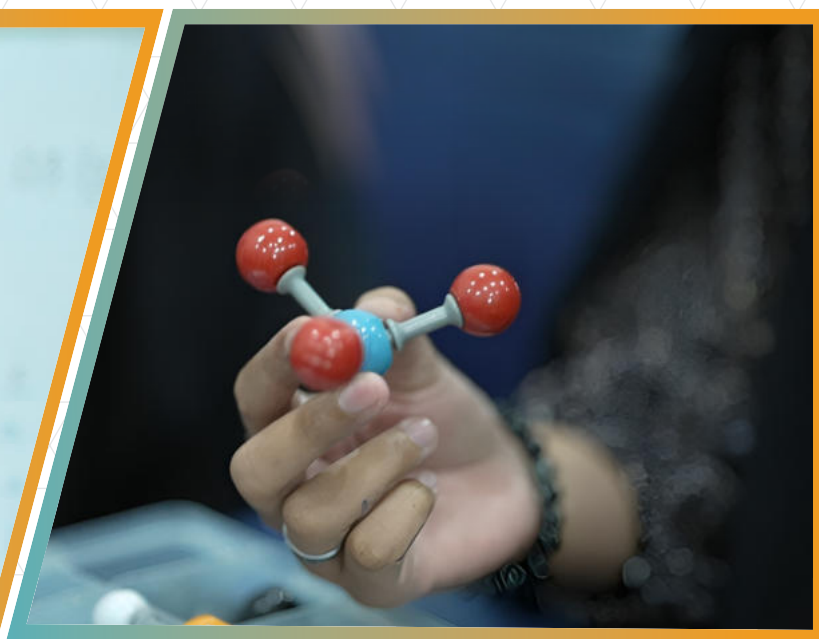
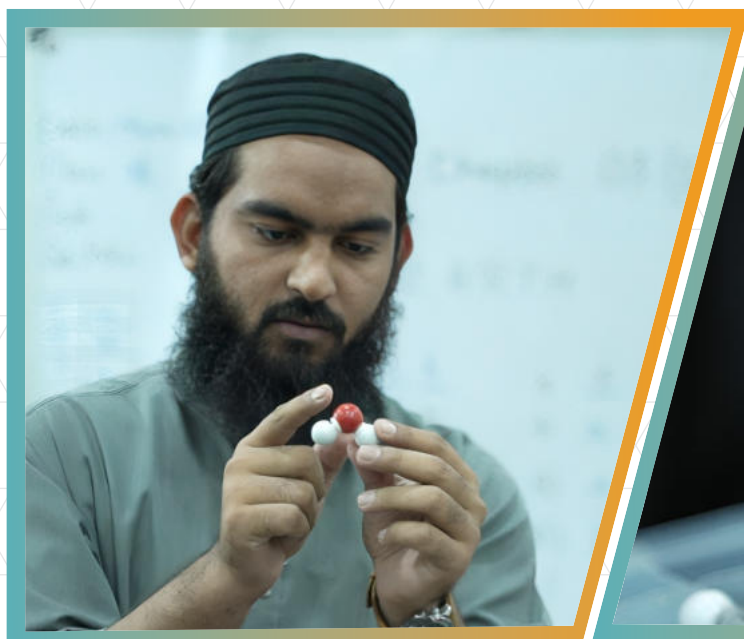




UNIT 04

CHEMICAL BONDING



WHY DO ATOMS FORM CHEMICAL BONDS?

It is a universal rule that everything in this world tends to become more stable. Atoms achieve stability by attaining electronic configuration of noble gases (He, Ne or Ar, etc) i.e. ns^2, np^6 . Having 2 or 8 electrons in the valence shell is sign of stability. Attaining two electrons in the valence shell is called duplet rule while attaining eight electrons in the valence shell is called octet rule.

SHORT QUESTIONS**1. Define chemical bond?****DEFINITION:**

"The force that holds atom together in a molecule."

2. Describe the concept of noble gas electronic configuration? Or for what reason atoms want to complete their valence shell?

Noble gases (He, Ne, Ar, Kr, Xe, Rn) are the most stable elements because they have full electron shells ($ns^2 np^6$, except Helium with $1s^2$). Other elements try to gain this stable electron configuration by filling their outer shells.

In order to achieve the stability atoms do chemical bonding.

3. What is octet rule? How many ways by which atoms complete their octet/duplet?

The tendency of atoms to make the number of their valence electrons eight, like the noble gases, is known as the octet rule. There are two ways for the elements to gain their octet and obtain a noble gas electron configuration.

1. Electron transfer

2. Electron sharing

4. What are valence electrons?

The number of electrons present in the outermost cloud/shell/orbit of an atom is known as valence electrons.

5. Describe the ways in which bonds may be formed?

The following may be the possibilities by which an atom forming the bond by another atom.

- ▶ Gaining or losing the valence electrons.
- ▶ Sharing the valence electrons.
- ▶ Donating or accepting the valence electrons.

6. How many types of chemical bonds?

Generally, three types of bonds depending upon the tendency of an atom.

- (i) Ionic bond.
- (ii) Covalent bond.
- (iii) Coordinate covalent bond.

2) Draw the concept of chemical species in chemistry?**IONS**

It is an atom or group of an atom having a charge on it. They consist of two types:

- I) Cation

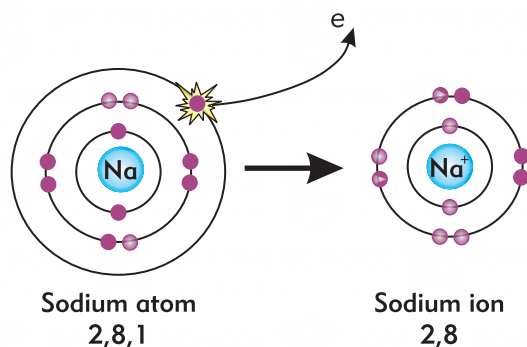
ii) Anion

i) CATION

The cations are formed when an atom loses electrons from outermost shell and hence they possess the positive electric charge on it (which shows that the number of protons is greater than the number of electrons)

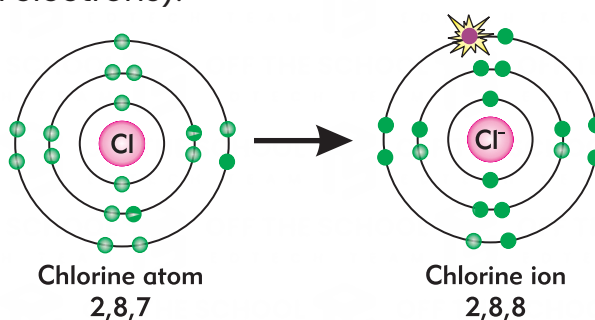


The formation of a sodium ion can be represented by a diagram as follows:



ii) ANION

The anion is formed when an atom gains electron in the outermost shell and hence they possess the negative electric charge on it (which shows that the number of protons is lesser than the number of electrons).



The formation of a chloride ion can be represented by a diagram as above:

MOLECULAR IONS

When a molecule loses or gains the electrons called molecular ions. Molecular ions also possess positive or negative charge like any ion. If it has negative charge known as anionic molecular ions, if they have positive charge known as cationic molecular ions. **For example:** CH_4^+ , H_2^+ , H_3O^+ , NH_4^+ all are anionic molecular ion and O_2^{2-} , N_2^{2-} are the common examples of cationic molecular ion.

FREE RADICALS

Free radicals are atoms and group of atoms having number of unpaired electrons. It is represented by putting a dot over the symbol of an element. **For example:** $\text{H}^\bullet$, $\text{Cl}^\bullet$, $\text{H}_3\text{C}^\bullet$.

LONG QUESTIONS

IONIC BONDS OR ELECTROVALENT BOND:

DEFINITION: Ionic bonds are formed by the complete transfer of electrons from one atom to another.

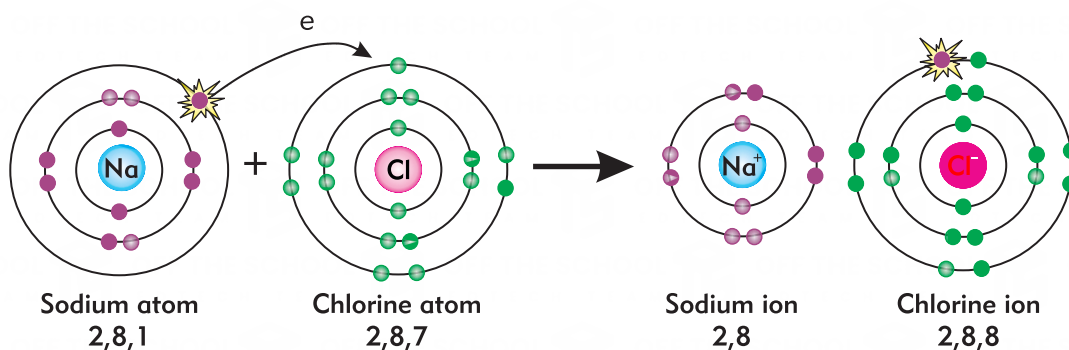
FORMATION OF IONIC BOND BETWEEN SODIUM AND CHLORINE ATOM:

The bond formation between sodium and chlorine, a metal and a nonmetal. The electronegativity values of sodium and chlorine are 0.9 and 3.0 respectively. This tells us that sodium has a low ionization energy and a tendency to give electrons while chlorine has a tendency to take electrons. When those two atoms come together under suitable conditions, to complete their octets, sodium gives one electron to chlorine.

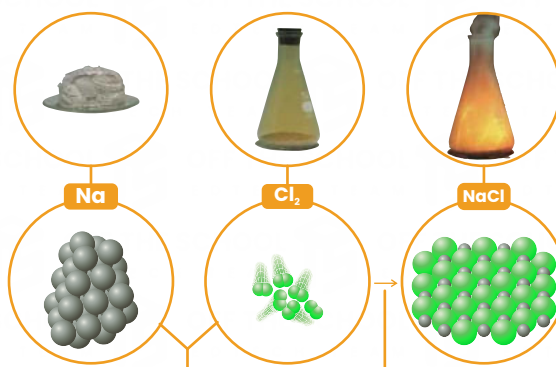
$_{11}\text{Na}: 1s^2, 2s^2, 2p^6, 3s^1$

$_{17}\text{Cl}: 1s^2, 2s^2, 2p^6, 3s^2, 3p^5$

Sodium loses its valence electron and its electron configuration becomes identical to that of neon: $1s^2 2s^2 2p^6$. Likewise, the valence shell of chlorine becomes completely filled and its electron configuration resembles that of argon: $1s^2, 2s^2, 2p^6, 3s^2, 3p^6$.



▲ Formation of ionic bond in Sodium chloride



Sodium metal and chlorine gas.

The combination reaction of sodium and chlorine. This reaction releases significant amounts of energy.

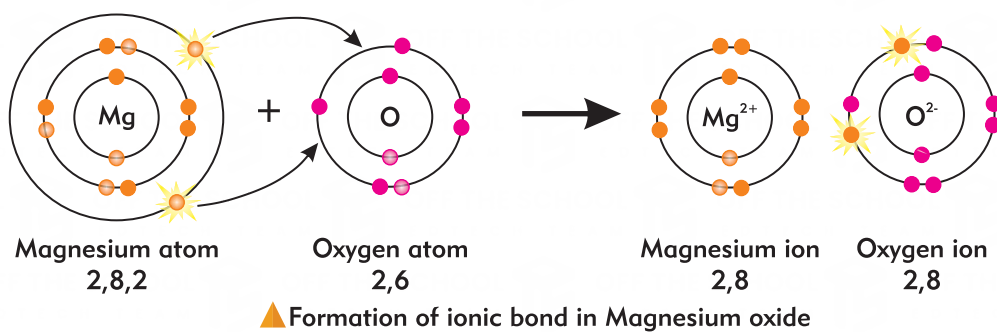
FORMATION OF MAGNESIUM OXIDE:

- ▶ Electronic configuration of Magnesium ($Z=12$) is $1s^2 2s^2 2p^6 3s^2$.
- ▶ Electronic configuration of Oxygen ($Z=8$) is $1s^2 2s^2 2p^4$.
- ▶ Magnesium loses two electrons from its valence shell while oxygen gains these two electrons such that both the elements attain the nearest inert gas configuration of neon.

Previous Year's Questions

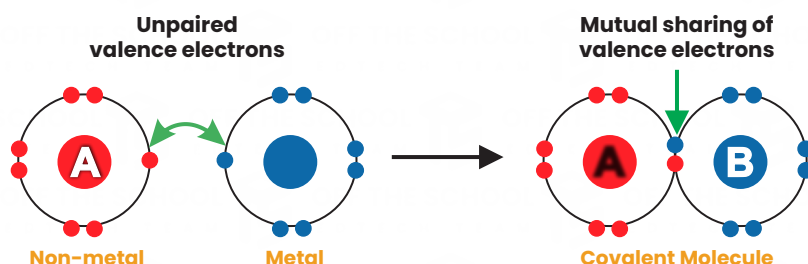
Define ionic bond. Illustrate the bond formation in sodium chloride (NaCl) also write any two characteristics of ionic compounds.

[2022,24]



COVALENT BOND:

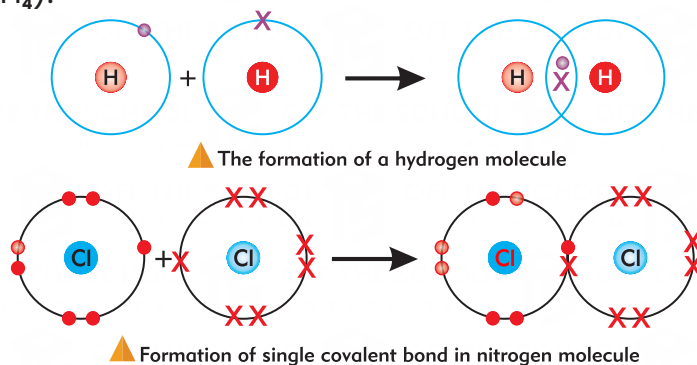
Covalent bond is a bond formed when two atoms share one or more electron pairs. Each atom contributes equal number of electrons towards the bond formation.



TYPES OF COVALENT BONDS:

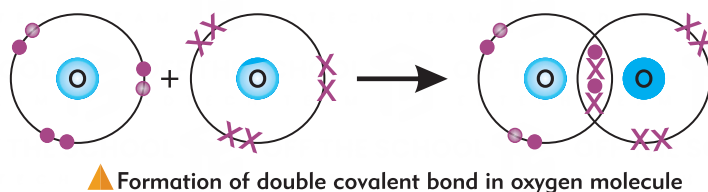
The covalent bond is formed by mutual sharing of electrons between two atoms. The electrons that pair up to form a chemical bond are called 'bond pair' electrons. Depending upon the number of bond pairs, covalent bond is classified into following three types:

I. SINGLE COVALENT BOND: When one electron is contributed by each bonded atom, one bond pair is formed and it forms a single covalent bond. A few examples of molecules with single covalent bonds are hydrogen (H_2), chlorine (Cl_2), hydrochloric acid (HCl) & methane (CH_4).

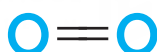


II. DOUBLE COVALENT BOND: When each bonded atom contributes two electrons, two bond pairs are shared and a double covalent bond is formed.

These bond pairs are indicated as double line between those atoms in the structure of such molecules. The molecules like oxygen (O_2) gas and ethene (C_2H_4) show such type of double covalent bonds.



The structural formula of an oxygen molecule is written

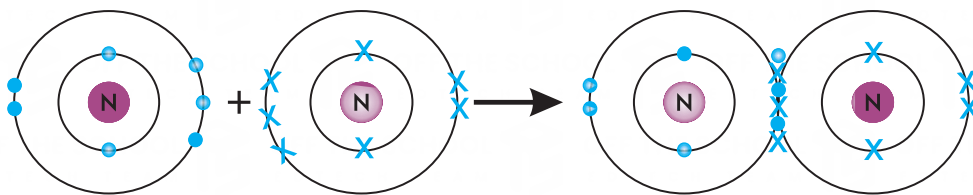


Previous Year's Questions

Why do some molecules form covalent bond? Write its three types with examples.

[2025]

III. TRIPLE COVALENT BOND: When each bonded atom contributes three electrons, three bond pairs are involved in bond formation. This type is called triple covalent bond.



▲ Formation of triple covalent bond in nitrogen molecule

The structural formula of a nitrogen molecule is:



Three small lines are used to indicate these three pairs of electrons between those atoms in the molecules. The examples of molecules having triple covalent bonds are nitrogen (N_2) and ethyne (C_2H_2).

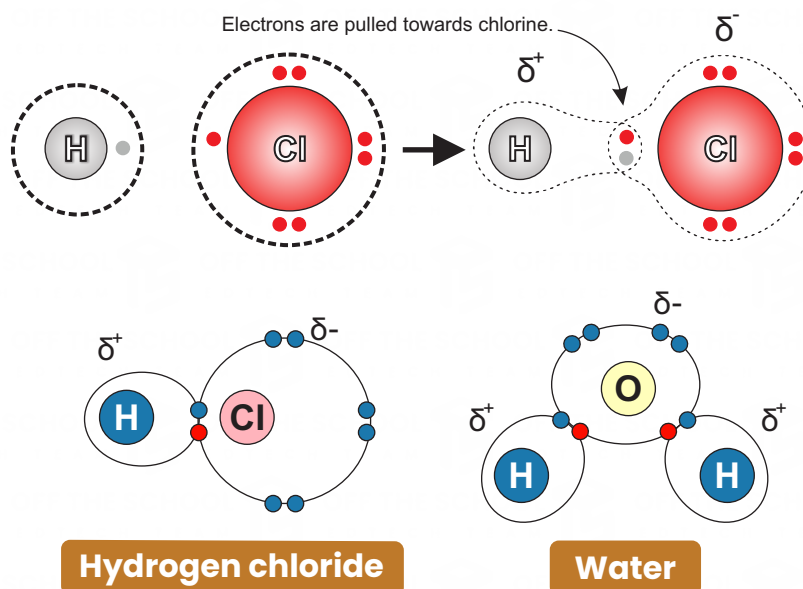
Student Note

The covalent bonds in all homonuclear diatomic molecules must be nonpolar like N_2 & O_2 .

POLAR COVALENT BOND:

When covalent bond is formed between two dissimilar atoms.

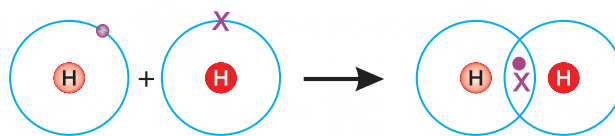
Polar covalent bond, characterized by a partial positive charge on one atom and a partial negative charge on the other. For example, the electrons in the H–Cl bond of a hydrogen chloride molecule spend more time near the chlorine atom than near the hydrogen atom. Thus, in an HCl molecule, the chlorine atom carries a partial negative charge and the hydrogen atom has a partial positive charge.



NON-POLAR COVALENT BOND:

When covalent bond is formed between two similar atoms.

If the atoms that form a covalent bond are identical, as in H_2 , Cl_2 , and other diatomic molecules, then the electrons in the bond must be shared equally. We refer to this as a pure covalent bond or non-polar covalent bond. Electrons shared in pure covalent bonds have an equal probability of being near each nucleus.



▲ The formation of a hydrogen molecule

COORDINATE COVALENT BOND OR DATIVE BOND:

DEFINITION:

The bonds form between atoms by the donation or acceptance of lone pair of electron is called coordinate covalent or dative bond.

OR

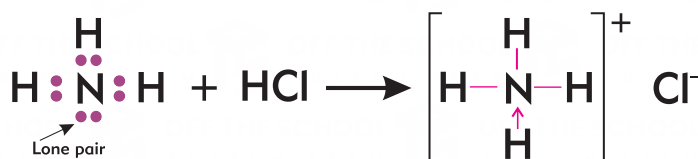
It may be defined as: a covalent bond in which both electrons of the shared pair come from one of the two atoms (or ions).

The atom A which donates the lone pair is called the donor, while B which accepts it the acceptor. The bond thus established is indicated by an arrow pointing from A to B. Although the arrow head indicates the origin of the electrons, once the coordinate bond is formed it is in no way different from an ordinary covalent bond.

EXAMPLE OF COORDINATE COMPOUNDS OR IONS:

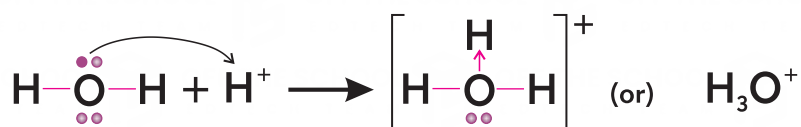
AMMONIUM ION, NH_4^+ :

In ammonia (NH_3), nitrogen is bonded to three hydrogen atoms and has a lone pair of electrons. A hydrogen ion (H^+) has no electrons, so it accepts the lone pair from nitrogen to form a coordinate bond, creating the ammonium ion (NH_4^+). All N-H bonds in NH_4^+ are identical after this bond forms.



▲ Reaction between Ammonia and Hydrogen Chloride

In water (H_2O), oxygen has lone pairs. One lone pair is donated to a hydrogen ion (H^+), forming H_3O^+ . All O-H bonds in H_3O^+ become equivalent.

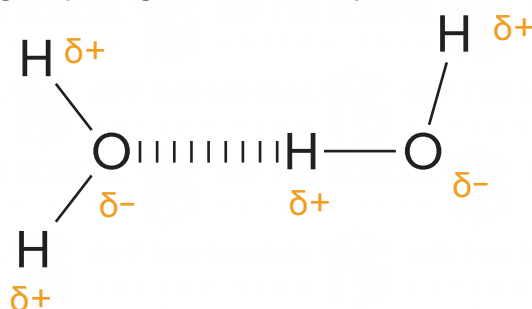


▲ Formation of Hydronium ion

HYDROGEN BONDING (H-Bonding):

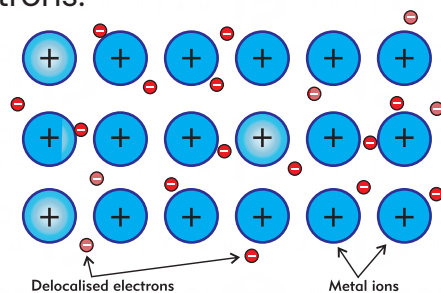
When hydrogen (H) is covalently bonded to a highly electronegative atom X (O, N, F), the shared electron pair is pulled so close to X this is hydrogen bonding.

DEFINITION: "The electrostatic attraction between an H atom covalently bonded to a highly electronegative atom X and a lone pair of electrons of X in another molecule, is called Hydrogen Bonding." Hydrogen bond is represented by a dashed or dotted line.



METALLIC BOND:

The metallic bond is defined as a bond formed between metal atoms (positively charged ions) due to mobile or free electrons.



▲ Diagrammatic representation of metallic bonding

PROPERTIES OF IONIC COMPOUNDS:

- (i) Ionic compounds are mostly crystalline solids at room temperature.
- (ii) Ionic compounds in solid state have negligible electrical conductance but they are good conductors in solution and in the molten form.
- (iii) Ionic compounds have high melting and boiling points.
- (iv) They dissolve easily in polar solvents like water.

PROPERTIES OF COVALENT COMPOUNDS:

- (i) They have usually low melting and boiling points.
- (ii) They are usually bad conductors of electricity.
- (iii) They are usually insoluble in water but are soluble in non-aqueous solvents like benzene, gasoline, alcohol and acetone.
- (iv) Large molecules of covalent crystals which are very stable and hard. They have very high melting and boiling points.

PROPERTIES OF POLAR COMPOUNDS:

- (i) They are soluble in water.
- (ii) They are usually conduct electricity.
- (iii) They are soluble in polar solvents.
- (iv) They are insoluble in non-polar solvents.

PROPERTIES OF NON-POLAR COMPOUNDS:

- (i) They are insoluble in water.
- (ii) They are usually do not conduct electricity.
- (iii) They are insoluble in polar solvents.
- (iv) They are soluble in non-polar solvents.

PROPERTIES OF COVALENT COMPOUNDS:

Their properties are mostly similar to those of covalent compounds.

PROPERTIES OF METALS:

- (i) All metals except mercury are solid.
- (ii) They have very high melting and boiling points.
- (iii) They have metallic lusture.
- (iv) Metals are generally ductile^[1] and malleable^[2].
- (v) When we hit the metal with hammer they produce sonorous.

WORKSHEET: Chemical Bonding – Ionic & Covalent!

REMEMBER...

Ionic Bond between a Metal and Non-Metal (M + NM)

Covalent Bond between a Non-Metal and Non-Metal (NM + NM)

PART 1: Determine if the elements in the following compounds are metals or non-metals. Describe the type of bonding that occurs in the compound.

Compound	Element 01 (metal or non-metal)	Element 02 (metal or non-metal)	Bond type
NO ₂	N = non metal	O = non metal	Covalent bond
NaCl			
SO ₂			
PI ₃			
MgBr ₂			
CaO			
H ₂ O			
K ₂ O			
AlF ₃			
O ₂			
CuCl ₂			
NO ₂			
CO ₂			
HF			
Rb ₂ S			
NBr ₃			
Fe ₂ O ₃			
CCl ₄			

PART 2: Use Lewis dot structures to show the ionic bonding in the following pairs of elements.

Show the transfer of electrons using arrows. Write the correct chemical formula for the ionic compound that forms.

1. Barium oxide (Ba and O)

Chemical formula: _____

Structure:

2. Sodium oxide (Na and O)

Chemical formula: _____

Structure:

3. Calcium chloride (Ca and Cl) Chemical formula: _____ Structure:	4. Sodium nitride (Na and N) Chemical formula: _____ Structure:
5. Aluminum oxide (Al and O) Chemical formula: _____ Structure:	6. Magnesium phosphide (Mg and P) Chemical formula: _____ Structure:

PART 3: Use Lewis dot structures to show the covalent bonding in the following pairs of elements; use a cross to represent a shared pair of electrons, and dots to show unshared electrons.

1) NITROGEN TRIIODIDE (NI_3)

Show work here... **HINT:** nitrogen is in the middle!

2) CARBON TETRABROMIDE (CBr_4)

Show work here... **HINT:** carbon is in the middle!